

2025 AMC 10A Problem 21 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 21. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 21

Problem:

A set of numbers is called sum-free if whenever x and y are (not necessarily distinct) elements of the set, $x + y$ is not an element of the set. For example, $\{1, 4, 6\}$ and the empty set are sum-free, but $\{2, 4, 5\}$ is not. What is the greatest possible number of elements in a sum-free subset of $\{1, 2, 3, \dots, 20\}$?

Answer Choices:

- A. 8
- B. 9
- C. 10
- D. 11
- E. 12

Solution:

Note that there are multiple solutions that lead to a set with size 10 - two examples are the odd numbers $\{1, 3, 5, \dots, 19\}$ and $\{11, 12, \dots, 20\}$.

We aim to prove that 11 elements is impossible. Let the largest element of our subset S be k , and consider the ordered pairs $(1, k - 1), (2, k - 2), \dots, (\lfloor \frac{k}{2} \rfloor, \lceil \frac{k}{2} \rceil)$.

Note that both elements of a pair may not be in S , as they would add to k . Therefore, we have at most $\lfloor \frac{k}{2} \rfloor \leq \frac{20}{2} = \boxed{\text{(C) } 10}$ elements, as desired.

Remark. This is a very famous problem in combinatorics, namely the *sum-free subset problem*. It was proven by Erdős that any set S of size n has a sum-free subset of size at least $\frac{n-1}{3}$.

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